CareConnect EHR System — Relational Database Specification

Author / Student: Utkarsh Gupta (Roll No: 2300320150065) | **Branch:** Computer Engineering
College: ABES ENGINEERING COLLEGE GHAZIABAD UTTAR PRADESH | **Email:** utkarsh.23b0151019@abes.ac.in
GitHub Repository: https://github.com/scrollingreel/P_003-CareConnect-Patient-EHR-HCL-TECH-PROJECT
Project Code: EHRS01 | **Engine:** Relational Database (H2 PostgreSQL Mode / PostgreSQL 16)
Migrations: Flyway V1 (Initial Schema), V2 (Departments & Doctors Seed), V3 (HCL EHRS01 Extended Entities)
Core Architecture: 17 Relational Tables, 4 Materialized Views, Foreign Key Integrity, ACID Transactions

1. Entity Alignment Matrix & Schema Tables

Table Name	HCL Reference Entity	Primary Key	Foreign Keys / Relationships	Purpose
`app_users`	Users & Roles	**id (UUID)**	Unique email constraint	Root credentials, BCrypt password hash, user roles (PATIENT, DOCTOR, ADMIN)
`patient_profiles`	Patients, MedicalHistory, Insurance	**id (VARCHAR 50)**	**user_id -> app_users(id)**	Demographics, insurance policy details, JSON allergies & chronic conditions
`doctors`	Providers	**id (VARCHAR 50)**	**department_id -> departments(id)**	Provider specialization, license number, consultation rooms
`clinical_notes`	EncounterNotes	**id (VARCHAR 50)**	**patient_id -> patient_profiles(id)**	Multi-specialty progress notes, separate assessment/plan, is_finalized lock
`lab_orders`	LabOrders (CPOE)	**id (VARCHAR 50)**	**patient_id, doctor_id**	LOINC laboratory test orders, urgency triage (Routine, Urgent, STAT)
`lab_results`	LabResults	**id (VARCHAR 50)**	**lab_order_id -> lab_orders(id)**	Quantitative values, reference ranges, abnormal and critical panic flags
`imaging_orders`	ImagingOrders (CPOE)	**id (VARCHAR 50)**	**patient_id, doctor_id**	Radiology requests (MRI, CT, Ultrasound) with anatomical body site
`imaging_results`	ImagingResults	**id (VARCHAR 50)**	**imaging_order_id -> imaging_orders(id)**	Radiologist diagnostic impression, findings narrative, DICOM accession
`procedure_orders`	ProcedureOrders (CPOE)	**id (VARCHAR 50)**	**patient_id, doctor_id**	Surgical and interventional procedures scheduling

Table Name	HCL Reference Entity	Primary Key	Foreign Keys / Relationships	Purpose
prescriptions	Prescriptions	id (VARCHAR 50)	patient_id, doctor_id	Electronic prescriptions with standardized drug dosage, route, and frequency
medication_administrations	MedicationAdministrations	id (VARCHAR 50)	prescription_id -> prescriptions(id)	Nurse-verified MAR administration logs (GIVEN, HELD, REFUSED)
medications_catalog	Medications	id (VARCHAR 50)	RxNorm code indexed	Master hospital pharmacology formulary catalog
secure_messages	SecureMessages (Patient Portal)	id (VARCHAR 50)	sender_id, recipient_id -> app_users(id)	Confidential patient-provider communication threads
patient_notifications	Notifications	id (VARCHAR 50)	patient_id -> patient_profiles(id)	Real-time clinical alerts, critical lab panic notifications, appointment alerts
audit_logs	AuditLogs	id (VARCHAR 50)	Indexed timestamp and user_id	Immutable tamper-evident record of all clinical data mutations

2. Database Views Specification

1. PatientSummary (`view_patient_summary`)

```
CREATE OR REPLACE VIEW view_patient_summary AS
SELECT
    p.id AS patient_id,
    p.user_id,
    u.name AS patient_name,
    u.email,
    p.dob,
    p.gender,
    p.blood_group,
    p.insurance_provider,
    p.insurance_policy_number,
    COUNT(DISTINCT rx.id) AS active_prescriptions_count,
    COUNT(DISTINCT lo.id) AS pending_lab_orders_count
FROM patient_profiles p
JOIN app_users u ON p.user_id = u.id
LEFT JOIN prescriptions rx ON rx.patient_id = p.id AND rx.status = 'ACTIVE'
LEFT JOIN lab_orders lo ON lo.patient_id = p.id AND lo.status IN ('PLACED', 'IN_PROGRESS')
GROUP BY p.id, p.user_id, u.name, u.email, p.dob, p.gender, p.blood_group, p.insurance_provider, p.insurance_policy_number;
```

2. LabOrdersAndResults (`view_lab_orders_and_results`)

```
CREATE OR REPLACE VIEW view_lab_orders_and_results AS
SELECT
    lo.id AS order_id,
    lo.patient_id,
    lo.test_name,
    lo.test_code,
    lo.urgency,
    lo.status AS order_status,
```

```
    lo.order_date,
    lr.id AS result_id,
    lr.result_value,
    lr.units,
    lr.reference_range,
    lr.status AS result_status,
    lr.is_critical
FROM lab_orders lo
LEFT JOIN lab_results lr ON lr.lab_order_id = lo.id;
```